PROJECT FOUNDATION DOSSIER

NetSentinel AI

Local-First Network Observability, Cybersecurity & Wireless Diagnostics Platform

Version: MVP / Project Foundation Dossier

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Classification: Confidential - Internal Project and Partner Evaluation

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1. Executive Summary

NetSentinel AI is a local-first network management, observability, cybersecurity, and outdoor wireless radio diagnostics platform. It is designed to help organizations monitor infrastructure, manage assets, detect alerts and incidents, analyze logs, run discovery scans, maintain security rules and indicators of compromise, automate response playbooks, and diagnose outdoor wireless radio links using real RF measurement data.

The project currently operates as an MVP / work in progress. Its runtime model is local deployment with Docker Compose, supported by a FastAPI backend, Next.js frontend, PostgreSQL database, Redis queue/cache layer, ARQ background worker, Electron desktop shell, and a Python-based edge agent for telemetry and wireless metric collection.

NetSentinel AI exists because many organizations still operate networks through fragmented tools: one system for assets, another for logs, another for alerts, spreadsheets for incidents, separate vendor portals for wireless devices, and informal technician notes for RF diagnostics. This fragmentation slows response, hides root causes, and makes operational knowledge hard to preserve.

The platform is intended for network administrators, Wireless ISP teams, cybersecurity analysts, IT managers, field technicians, small enterprises, municipalities, and public institutions that need practical local infrastructure visibility without immediately adopting a complex cloud SIEM, enterprise NMS, or multiple disconnected vendor tools.

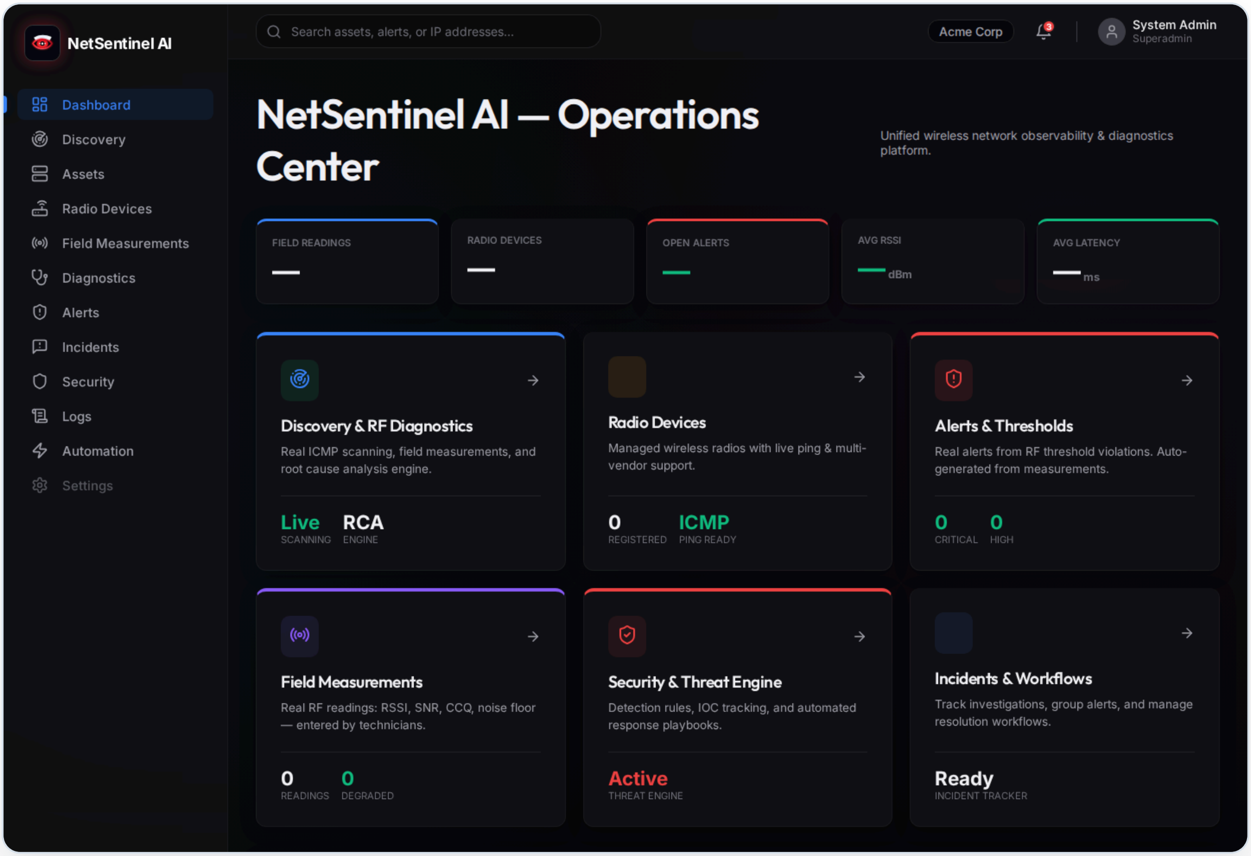
NetSentinel AI differs from normal monitoring tools by combining network operations, security monitoring, field workflows, AI-assisted explanation, and outdoor wireless diagnostics in a single operational environment. Outdoor wireless diagnostics are especially important because RF problems are frequently physical, environmental, intermittent, and difficult to interpret from raw numbers alone. Metrics such as RSSI, SNR, noise floor, CCQ, latency, packet loss, frequency, channel width, antenna alignment, and Fresnel zone clearance must be interpreted together.

MVP LIMITATION

NetSentinel AI should not currently be described as production-ready. The codebase shows a runnable foundation, but additional validation, tests, security hardening, RBAC, audit logging, rate limiting, deployment hardening, and real integrations are required before enterprise production use.

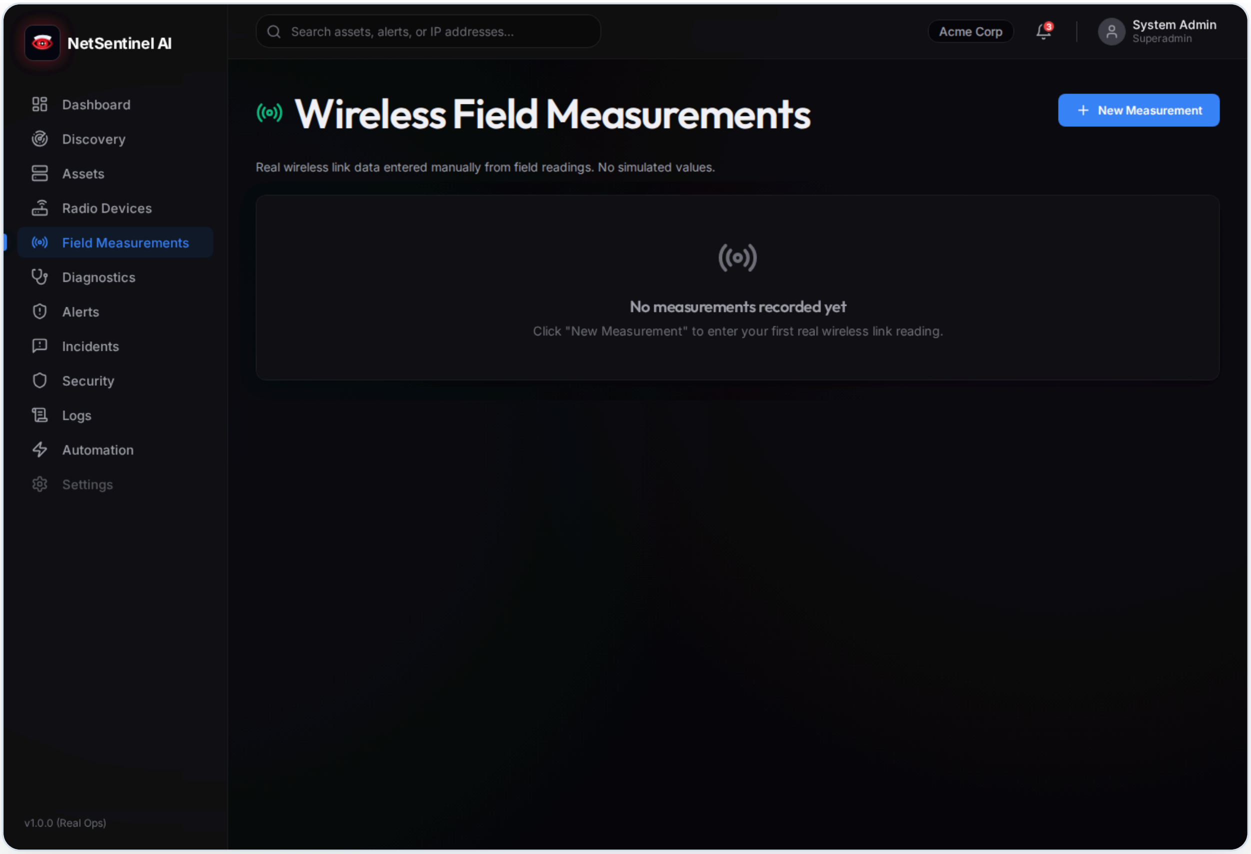
2. Product Interface Preview

The following interface captures show the current MVP visual direction. They demonstrate the desktop-style operations experience and the field workflow surface for wireless measurements. These screens should be understood as MVP / work-in-progress views, not final production claims.



Operations Center dashboard showing the dark NOC-style navigation, operational cards, alert counters, discovery entry points, and module shortcuts.

Figure: Operations Center Dashboard. This screen demonstrates the unified operations landing page, including navigation to discovery, assets, radio devices, field measurements, alerts, incidents, security, logs, and automation.



Wireless Field Measurements page showing the technician-facing RF measurement surface and empty-state behavior when no measurements are recorded.

Figure: Wireless Field Measurements. This screen demonstrates the field data entry workflow for real wireless link readings such as RSSI, SNR, CCQ, noise floor, packet loss, and latency.

3. Project Vision

The long-term vision for NetSentinel AI is to become a local-first NOC/SOC platform for organizations that need operational control, network visibility, security awareness, and field-grade wireless diagnostics without surrendering all operational data to external cloud services.

The platform vision includes:

Area	Vision
Local-first operations	Run core infrastructure monitoring locally, on-premise, or inside controlled private environments.
Network visibility	Maintain a unified inventory of organizations, sites, assets, wireless links, measurements, alerts, incidents, and logs.
Cybersecurity monitoring	Detect suspicious events using rules, IOCs, logs, and correlated alerts.
Wireless RF diagnostics	Translate raw RF measurements into technician-ready diagnostic insight.
AI-assisted troubleshooting	Explain incidents, summarize logs, propose likely causes, and recommend next steps based on internal evidence.
Automation	Use playbooks and response actions to standardize incident handling while preserving human approval.
Desktop-first experience	Provide a familiar local desktop launcher for operators who prefer an installed NOC/SOC workstation experience.

FUTURE IMPROVEMENT

The ideal mature version should support real-time updates, robust agent management, scheduled discovery, role-based dashboards, SIEM export, SNMP integrations, wireless vendor adapters, PDF reporting, and compliance workflows.

4. Problem Statement

Network and security teams often operate under pressure with incomplete or disconnected information. In many small and medium organizations, the operator must switch between router interfaces, wireless controller portals, spreadsheets, manual notes, log files, ticket systems, and messaging channels to understand a single incident.

Key problems include:

Problem	Operational Impact
Fragmented tools	Slow troubleshooting and inconsistent incident records.
Disconnected assets, alerts, logs, and incidents	Root cause analysis becomes manual and error-prone.
Limited local NOC/SOC affordability	Smaller organizations cannot justify large enterprise platforms.
Difficult outdoor wireless diagnosis	RF problems require specialized knowledge and field context.
Weak historical context	Technician observations and measurements are not preserved in a structured way.
Limited automation	Teams repeat manual triage and response steps during recurring incidents.
Security blind spots	IOCs, rules, and logs may not be connected to operational assets and incidents.

Outdoor wireless environments add a specialized layer of difficulty. A link can degrade because of interference, antenna misalignment, cable damage, water ingress, vegetation growth, tower movement, Fresnel zone obstruction, thermal effects, frequency reuse, or asymmetric endpoint conditions. Raw signal values alone rarely tell the full story.

5. Proposed Solution

NetSentinel AI proposes a unified local platform that connects network operations, cybersecurity visibility, wireless diagnostics, and automation into one operational workspace.

Core solution components include:

Component	Contribution
Centralized dashboard	Provides a single operations view for assets, alerts, incidents, wireless links, logs, and security posture.
Asset inventory	Maintains structured records of network devices, servers, endpoints, and related sites.
Discovery scans	Performs network scan workflows and stores discovered host results.
Alerts and incidents	Converts operational or security findings into structured workflows.
Security rules and IOCs	Supports security detection concepts and future threat correlation.
Logs	Provides storage and review surfaces for event and system logs.
Automation playbooks	Supports future standardized response actions.
Edge telemetry agent	Collects telemetry and wireless metrics near the monitored environment.
Wireless measurement forms	Captures field measurements and RF diagnostics data.
AI assistant	Explains alerts, incidents, and wireless problems using internal evidence.
Docker deployment	Makes the MVP reproducible locally.
Desktop launcher	Gives operators a native desktop entry point into the local dashboard.

6. Target Users

Network Administrator

Responsibilities: Maintain uptime, manage infrastructure inventory, investigate connectivity problems, track device status, and coordinate changes.

Needs: Clear asset records, alerts, discovery scans, logs, topology context, and practical troubleshooting workflows.

Wireless ISP Technician

Responsibilities: Install, align, maintain, and troubleshoot outdoor PTP/PTMP links and customer-premise radios.

Needs: RF metric history, field measurement capture, diagnostic scoring, likely-cause hints, maintenance logs, and technician action plans.

Cybersecurity Analyst

Responsibilities: Review suspicious events, correlate alerts, maintain IOCs, investigate incidents, and recommend remediation.

Needs: Rules, IOCs, logs, incidents, evidence trails, risk scoring, and safe automation controls.

IT Manager

Responsibilities: Oversee service quality, team performance, incident response, risk management, and reporting.

Needs: Executive dashboard, status summaries, incident reports, SLA indicators, and roadmap confidence.

Field Technician

Responsibilities: Visit physical sites, inspect cabling, align antennas, replace equipment, and record measurements.

Needs: Mobile-friendly forms, pre-visit briefs, maintenance history, clear instructions, and offline-friendly workflows.

Small Enterprise Owner

Responsibilities: Maintain business continuity with limited IT staff and limited budget.

Needs: Local deployment, simple dashboard, affordable monitoring, alerts, and practical reports.

Municipality / Public Institution IT Team

Responsibilities: Maintain public networks, remote sites, surveillance links, civic infrastructure, and service continuity.

Needs: Local data control, auditability, operational visibility, wireless diagnostics, and future compliance reporting.

7. Core Platform Modules

Dashboard / Operations Center

Category	Description
Purpose	Provide a unified operational starting point for network, security, and wireless status.
Main features	Summary cards, navigation, recent alerts, incident visibility, asset context, dark operations UI.
Inputs	API data from assets, alerts, incidents, logs, security, wireless, discovery, and metrics.
Outputs	Operator awareness, navigation into investigation pages, high-level status indicators.
Current MVP state	Frontend dashboard structure exists. Data wiring varies by module.
Future improvements	Real-time updates, custom widgets, role-based views, SLA cards, topology map.

Discovery Engine

Category	Description
Purpose	Identify reachable hosts on local network subnets.
Main features	CIDR scan request, ICMP ping sweep, scan results, host import concept.
Inputs	Subnet/CIDR, scan parameters, authenticated user context.
Outputs	Discovery scan records and discovered host records.
Current MVP state	Backend includes real ICMP ping-based subnet scanning service.
Future improvements	Scheduled scans, port detection, OS fingerprinting, SNMP discovery, agent-based scanning.

Asset Inventory

Category	Description
Purpose	Maintain structured records for devices, servers, endpoints, and network assets.
Main features	List, create, update, delete, stats, site association.
Inputs	Asset metadata, site context, discovery imports.
Outputs	Asset records used by alerts, incidents, and dashboards.
Current MVP state	Backend models, schemas, routes, and frontend page exist.

Category	Description
Future improvements	Lifecycle tracking, ownership, warranty, firmware, credentials vault references, topology mapping.

Radio Devices

Category	Description
Purpose	Track physical radio devices used in wireless networks.
Main features	Device inventory, vendor fields, adapter type, role, ping operations, diagnostic entry points.
Inputs	Radio metadata, IP addresses, vendor, role, field measurements.
Outputs	Radio device records and diagnostic reports.
Current MVP state	Backend router and frontend page exist.
Future improvements	Vendor APIs, SNMP profiles, MikroTik/Ubiquiti/TP-Link adapters, credential profiles.

Field Measurements

Category	Description
Purpose	Capture technician observations and measured link conditions from the field.
Main features	Measurement records, status stats, update and delete operations.
Inputs	Signal values, latency, packet loss, capacity, notes, site/radio/link context.
Outputs	Structured field evidence for diagnostics and reports.
Current MVP state	Backend router and frontend page exist.
Future improvements	Mobile-first capture, photo attachments, GPS metadata, offline draft mode.

Wireless Diagnostics

Category	Description
Purpose	Interpret RF metrics and field evidence for outdoor wireless links.
Main features	Wireless links, metrics, diagnostics, maintenance logs, AI field brief endpoint.
Inputs	RSSI, SNR, noise floor, CCQ, latency, packet loss, capacity, frequency, channel width, notes.
Outputs	Health score, likely cause, technician action plan, alert candidates.
Current MVP state	Wireless models, routes, pages, metrics, diagnostics, and field brief concepts exist.

Category	Description
Future improvements	Deterministic scoring engine, vendor-specific thresholds, trend detection, spectrum analysis.

Alerts & Thresholds

Category	Description
Purpose	Surface conditions that require attention.
Main features	Alert CRUD, severity, status, stats, asset or wireless link association.
Inputs	Rules, thresholds, telemetry, security events, manual entries.
Outputs	Alerts linked to incidents and dashboards.
Current MVP state	Backend routes, schemas, models, and frontend page exist.
Future improvements	Rule builder, deduplication, suppression windows, notification channels.

Incidents & Workflows

Category	Description
Purpose	Group related alerts into investigation and response workflows.
Main features	Incident list, stats, status changes, severity, CRUD.
Inputs	Alerts, manual incident creation, AI analysis, operator updates.
Outputs	Incident records, workflow state, investigation history.
Current MVP state	Backend routes and frontend page exist.
Future improvements	Timeline, assignment, comments, SLA, escalation, post-incident report.

Security & Threat Engine

Category	Description
Purpose	Support security rules, IOCs, and future threat detection workflows.
Main features	Detection rules, IOCs, security page, threat engine service concepts.
Inputs	Logs, IOCs, rules, telemetry, asset context.
Outputs	Security alerts, incident candidates, risk indicators.
Current MVP state	Backend models/routes for rules and IOCs exist.
Future improvements	MITRE mapping, correlation engine, risk score, vulnerability integration.

Logs

Category	Description
Purpose	Store and expose log events for observability and investigation.
Main features	Log ingestion/listing, filters, frontend log page.
Inputs	Application events, device logs, future syslog, agent telemetry.
Outputs	Searchable operational evidence and security signals.
Current MVP state	Backend log routes and frontend page exist.
Future improvements	Syslog collector, full-text search, retention policies, export.

Automation

Category	Description
Purpose	Define response playbooks and track response actions.
Main features	List playbooks, list recent response actions.
Inputs	Alerts, incidents, operator approvals, playbook rules.
Outputs	Response action records and future remediation execution.
Current MVP state	Backend route is present; capabilities are early.
Future improvements	Action execution, approval workflow, rollback, integrations, audit log.

AI Assistant

Category	Description
Purpose	Explain alerts, incidents, logs, and wireless degradation in operational language.
Main features	Chat endpoint, incident analysis, alert analysis, capabilities endpoint, wireless field brief concept.
Inputs	User prompt, incident/alert IDs, internal context, optional Gemini API key.
Outputs	Explanation, remediation suggestions, root cause hypotheses, confidence metadata.
Current MVP state	Backend AI endpoints exist with Gemini/fallback behavior.
Future improvements	Evidence citations, guarded tool use, prompt injection defenses, audit trail.

Edge Agent

Category	Description
Purpose	Collect telemetry and wireless metrics near the monitored network.
Main features	SNMP client concept, wireless polling, backend transport, demo polling loop.
Inputs	Device IPs, community strings, vendor profile, backend URL, agent key.
Outputs	Normalized telemetry and wireless metrics sent to backend.
Current MVP state	Python agent entry point and polling structure exist.
Future improvements	Secure enrollment, offline queue, configuration sync, vendor adapters, health reporting.

Desktop Client

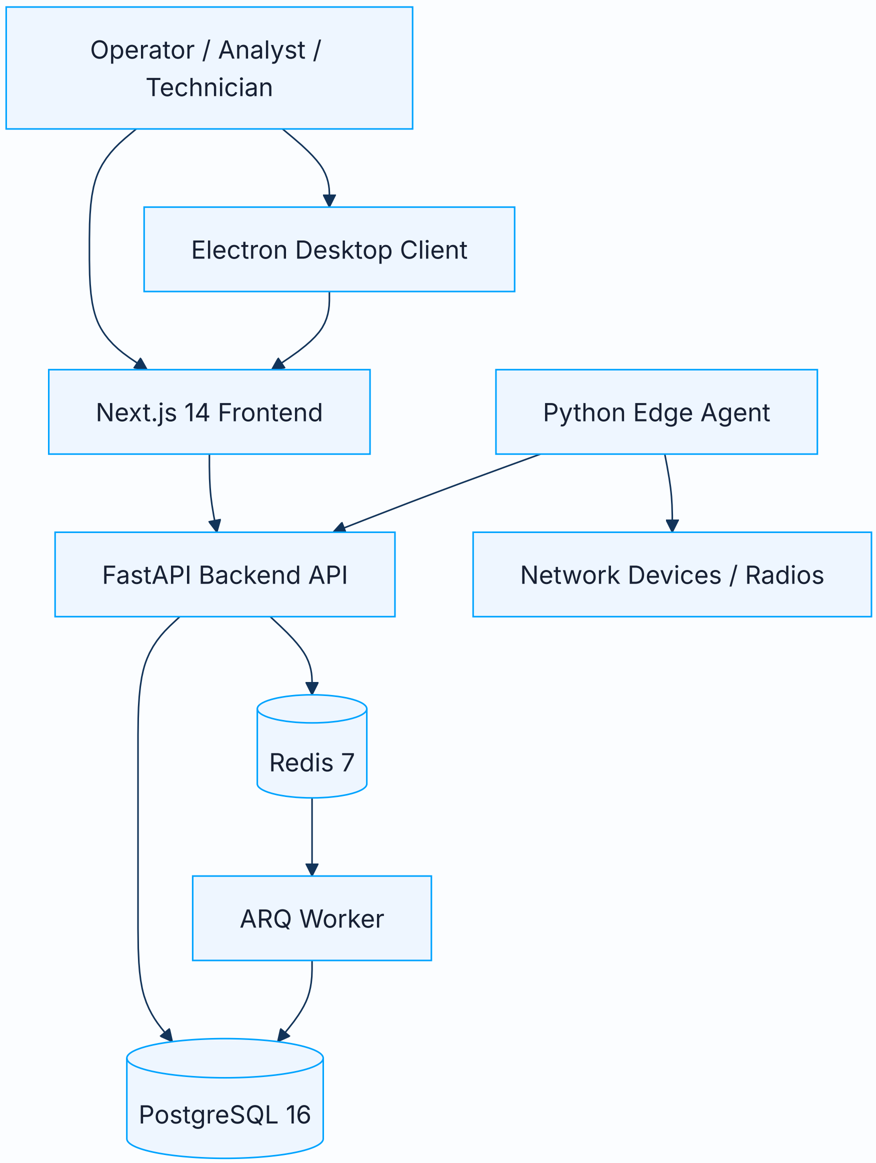
Category	Description
Purpose	Provide a native desktop shell for local operators.
Main features	Electron window opening local dashboard, Linux launcher script, desktop entry.
Inputs	Local frontend URL and Docker Compose service status.
Outputs	Desktop application experience.
Current MVP state	Electron client and launcher exist.
Future improvements	Tray icon, auto-start, local health indicator, packaging for Linux and Windows.

8. Technical Architecture

High-Level Architecture

NetSentinel AI follows a modular local-first architecture. The MVP is deployed as a Docker Compose stack with a web dashboard, API backend, relational database, queue/cache layer, worker process, and edge telemetry agent.

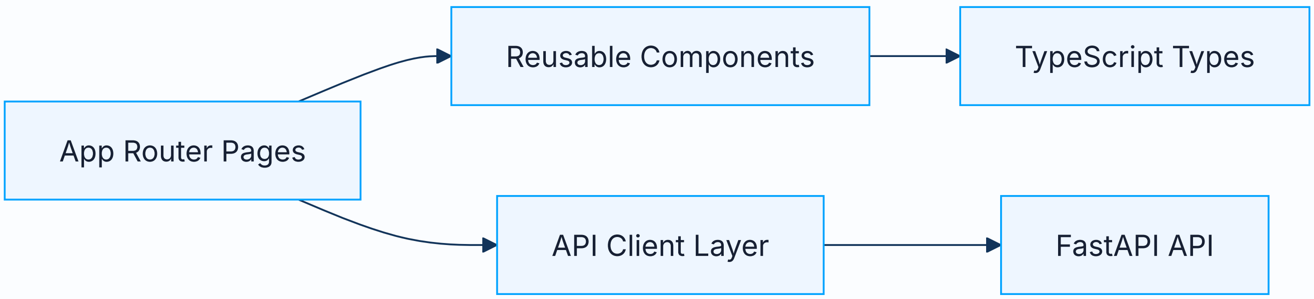
ARCHITECTURE DIAGRAM



Frontend Architecture

The frontend is a Next.js 14 application using TypeScript, React 18, CSS modules, SWR, Recharts, and lucide-react icons. It provides pages for the dashboard, assets, alerts, incidents, logs, security, automation, discovery, wireless, diagnostics, radio devices, and field measurements.

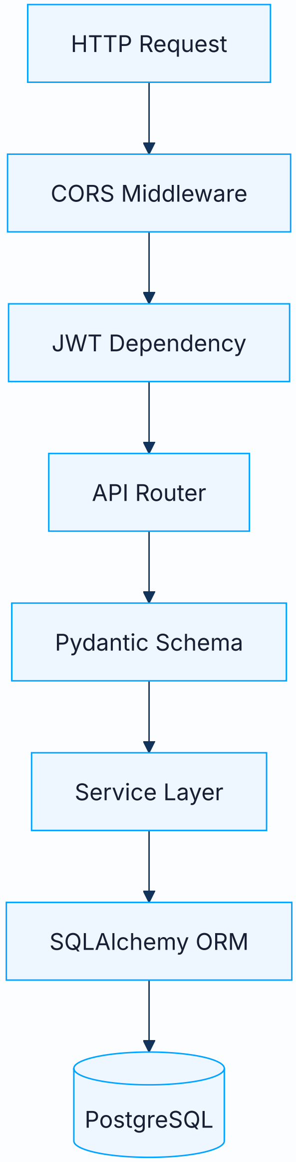
ARCHITECTURE DIAGRAM



Backend Architecture

The backend is a FastAPI application with routers, schemas, SQLAlchemy models, service modules, authentication dependencies, AI modules, ingestion endpoints, and worker integration.

ARCHITECTURE DIAGRAM



Database Architecture

PostgreSQL stores core operational data: organizations, users, sites, assets, alerts, incidents, logs, security rules, IOCs, radio devices, wireless links, metrics, field measurements, diagnostics, automation playbooks, and response actions. SQLAlchemy models define the persistence layer and Alembic is present for migrations.

Redis / Queue Architecture

Redis is used as a cache and queue foundation. The ARQ worker is wired through Redis and is intended to support future scheduled tasks, alert processing, telemetry processing, discovery jobs, notification jobs, and automation workflows.

Edge Agent Architecture

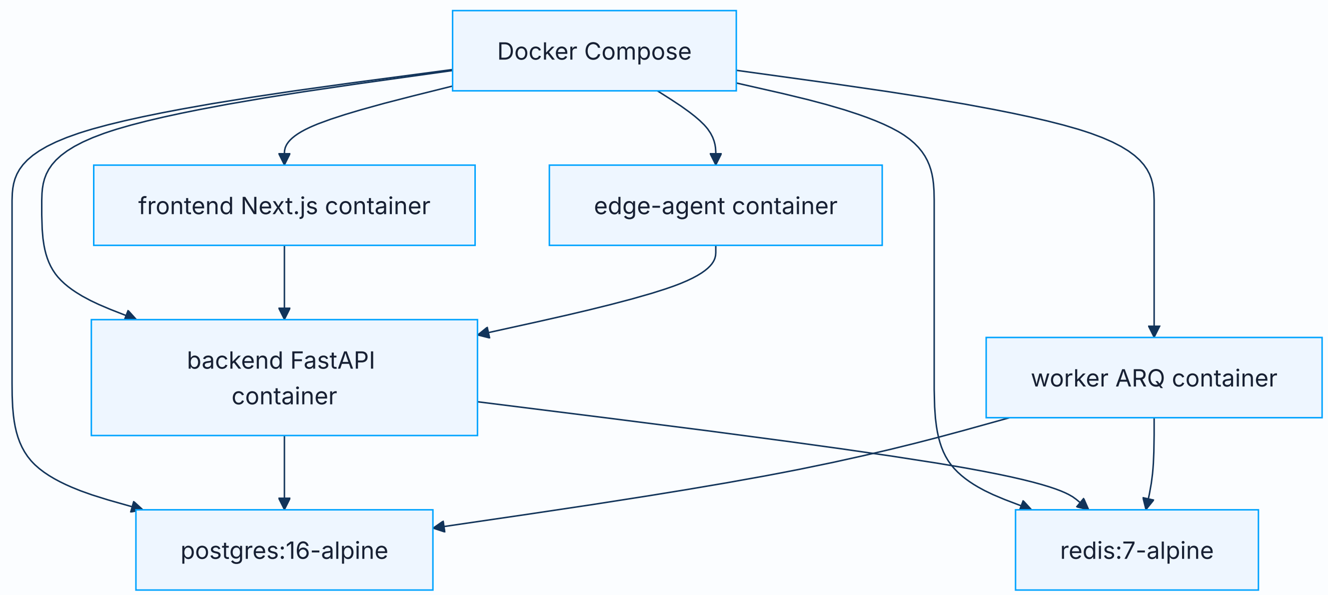
The edge agent is a standalone Python process intended to run near customer networks. It includes SNMP polling concepts, wireless metric collection, and backend transport. The current implementation includes demo wireless targets and a polling loop.

Desktop Client Architecture

The Electron desktop client wraps the local dashboard. The Linux launcher starts Docker Compose services, waits for frontend/backend readiness, and opens the local interface.

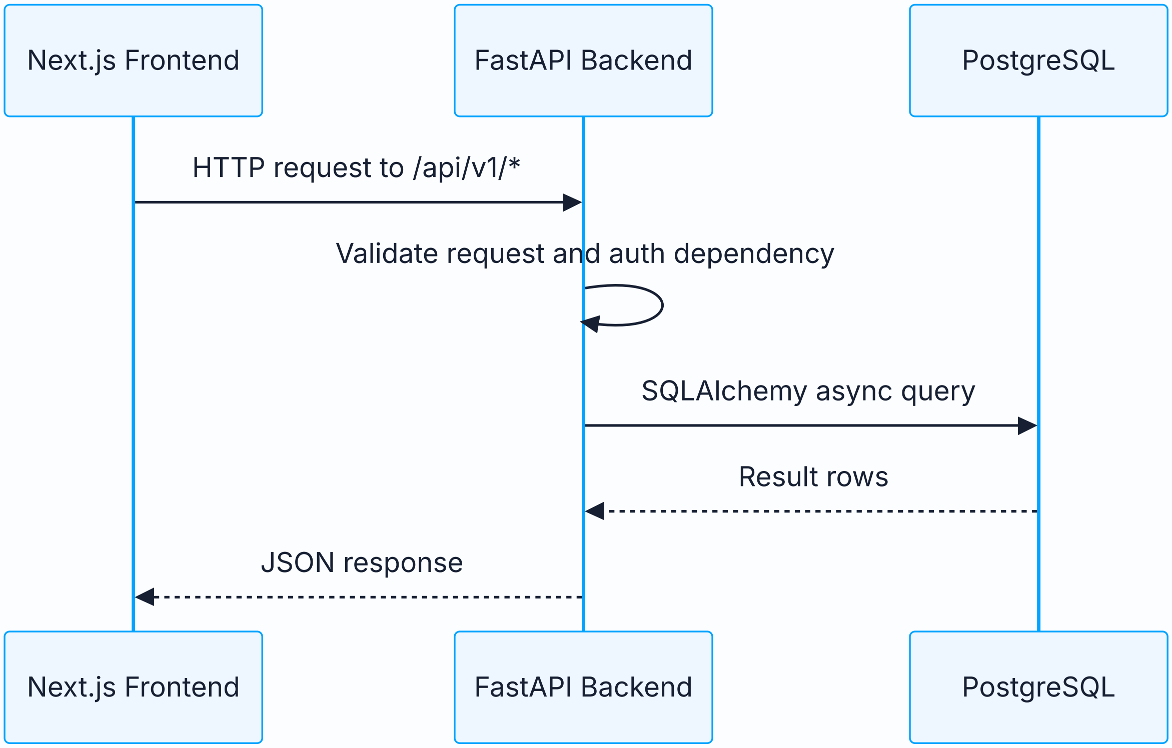
Docker Compose Architecture

ARCHITECTURE DIAGRAM



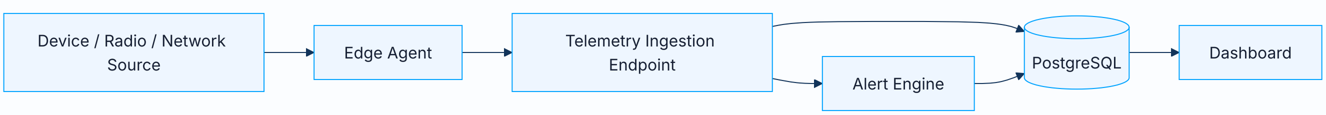
API Communication Flow

ARCHITECTURE DIAGRAM



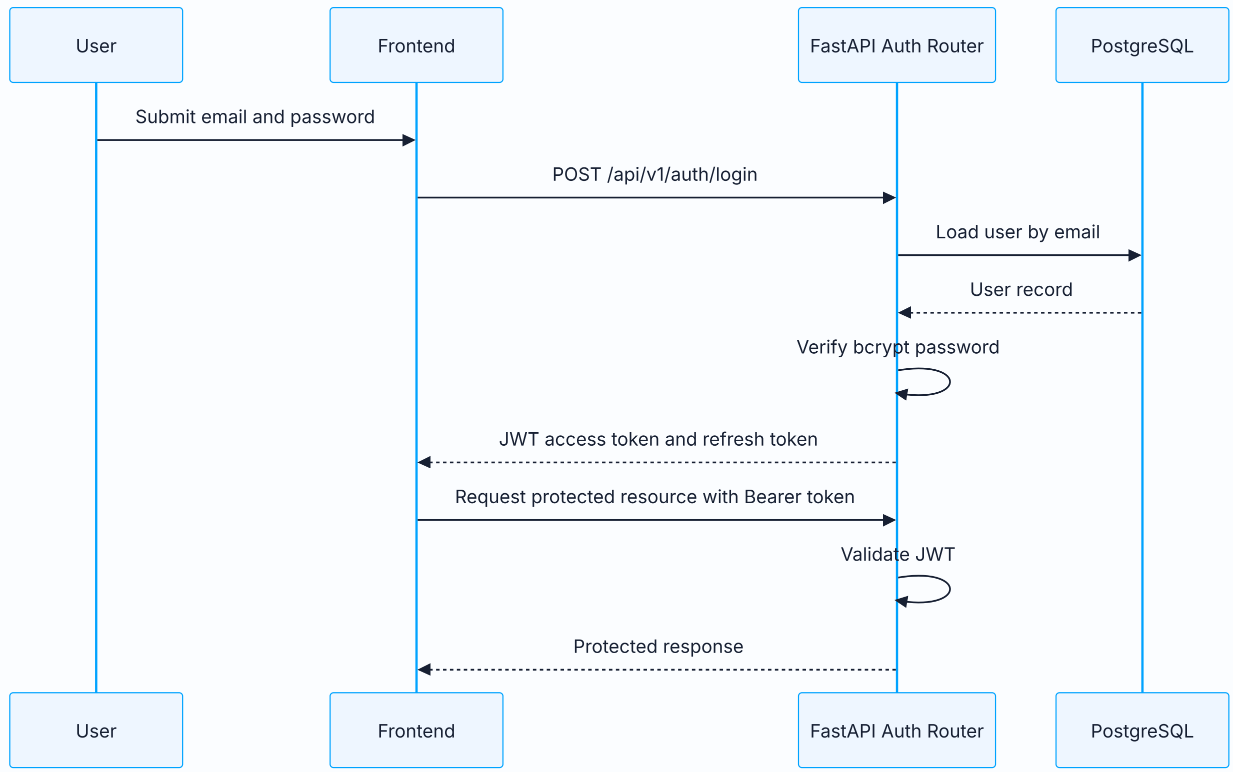
Data Ingestion Flow

ARCHITECTURE DIAGRAM



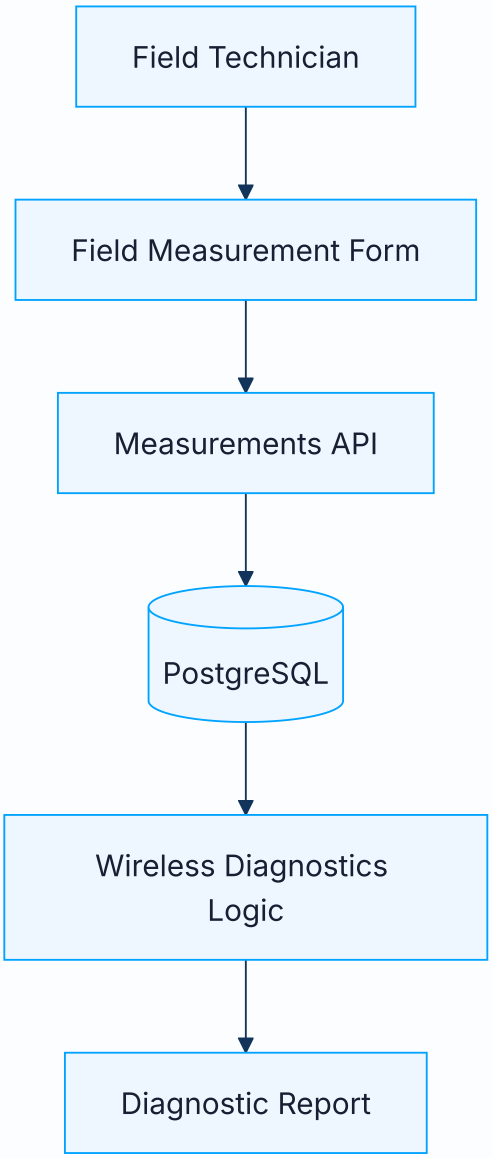
Authentication Flow

ARCHITECTURE DIAGRAM



Wireless Measurement Flow

ARCHITECTURE DIAGRAM



Alert Generation Flow

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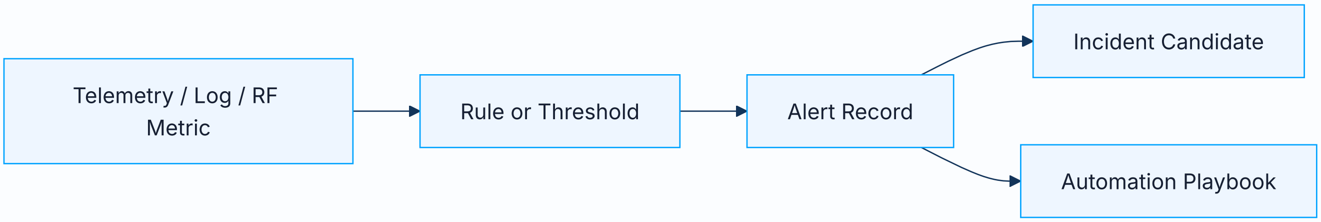

AI Assistant Flow

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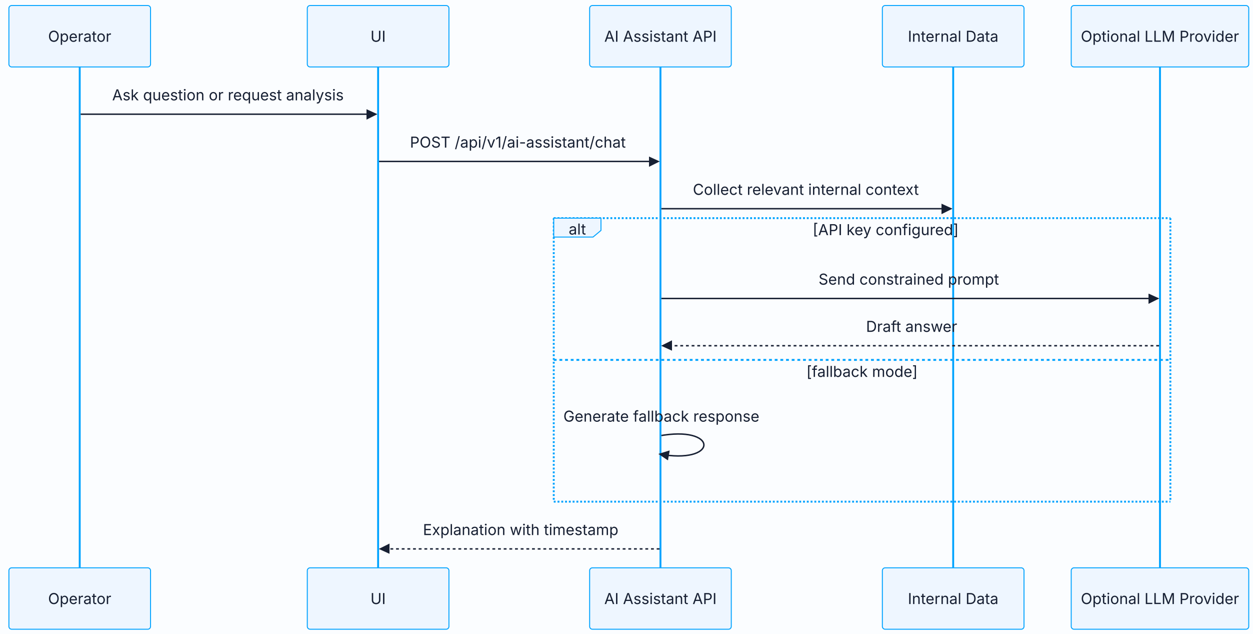
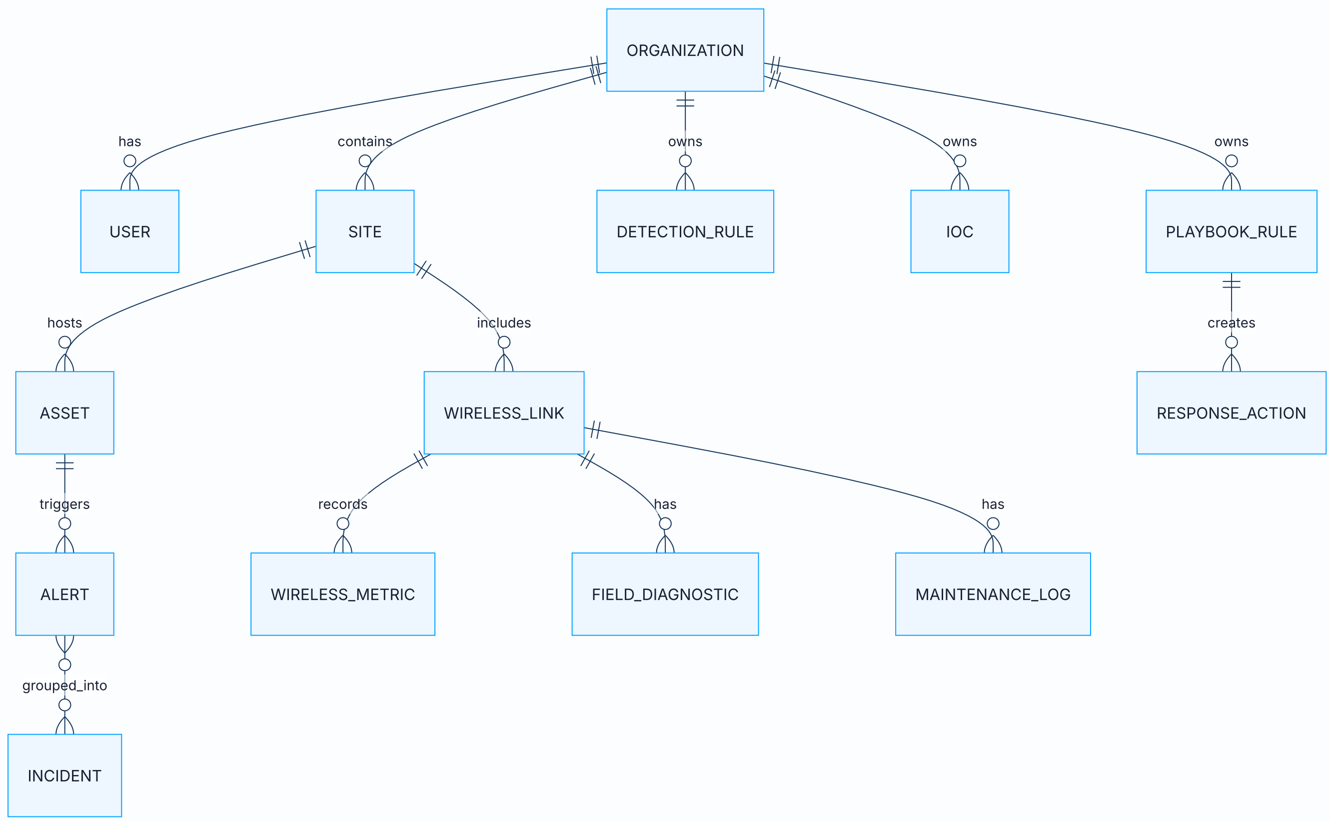


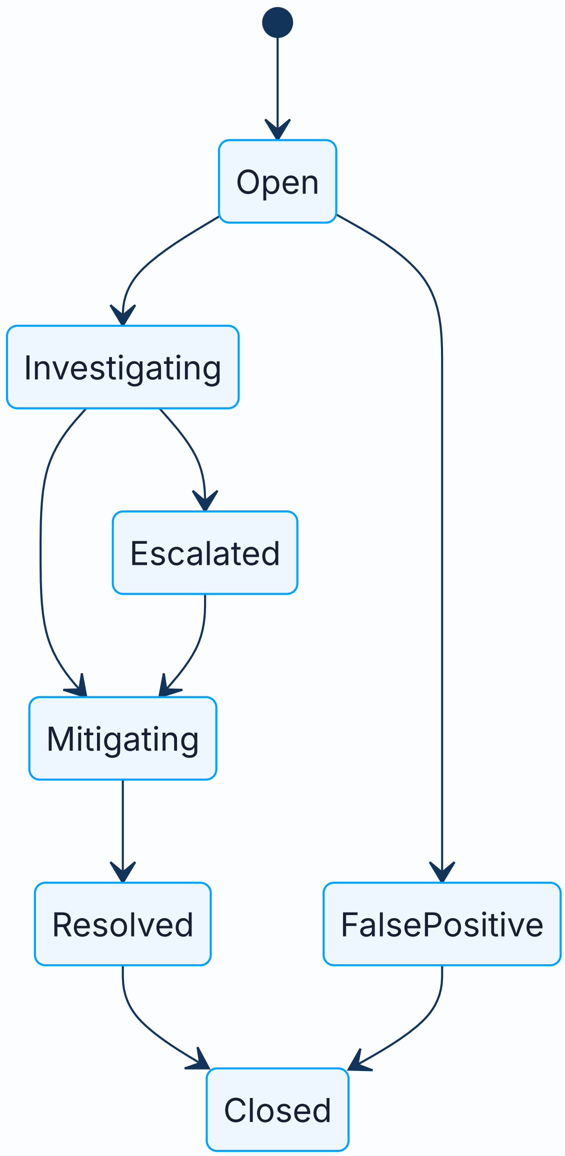
Conceptual Entity Relationship Diagram

ARCHITECTURE DIAGRAM



Incident Workflow Diagram

ARCHITECTURE DIAGRAM



9. Backend Documentation

The FastAPI backend is the authoritative API layer for NetSentinel AI. It exposes `/api/v1/*` routers, manages application startup, initializes database tables during MVP runtime, configures CORS, and provides a `/health` endpoint.

API Structure

Area	Endpoint Group	Role
Authentication	`/api/v1/auth/*`	Register, login, refresh, current user.
Organizations	`/api/v1/organizations/*`	Multi-organization management.
Sites	`/api/v1/sites/*`	Physical/logical site records.
Assets	`/api/v1/assets/*`	Device and infrastructure inventory.
Alerts	`/api/v1/alerts/*`	Alert lifecycle and stats.
Incidents	`/api/v1/incidents/*`	Incident workflow and stats.
Logs	`/api/v1/logs/*`	Log creation and listing.
Security	`/api/v1/security/*`	Detection rules and IOCs.
Automation	`/api/v1/automation/*`	Playbooks and response actions.
Discovery	`/api/v1/discovery/*`	Scans and discovered hosts.
Wireless	`/api/v1/wireless/*`	Links, metrics, diagnostics, maintenance.
Field Measurements	`/api/v1/field-measurements/*`	Field measurement records.
Radio Devices	`/api/v1/radio-devices/*`	Radio inventory and diagnostics.
AI Assistant	`/api/v1/ai-assistant/*`	Chat, alert analysis, incident analysis.
Telemetry	`/api/v1/metrics`	Telemetry ingestion endpoint.

Authentication and JWT

The backend uses JWT access and refresh tokens. Password hashing uses bcrypt. The MVP settings define access token and refresh token lifetimes through environment variables.

> **SECURITY NOTE**
>
> The default development secret key must be changed before any shared or production-like deployment. JWT signing secrets should be generated with high entropy and stored outside source control.

API Documentation Strategy

FastAPI automatically exposes interactive documentation:

Documentation	URL
Swagger UI	`http://localhost:8000/docs`
ReDoc	`http://localhost:8000/redoc`

Recommended additions:

Recommendation	Rationale
Add endpoint descriptions and examples	Improves onboarding for engineers and external reviewers.
Add error response models	Makes API behavior predictable.
Add authentication examples	Helps frontend and integrators use JWT correctly.
Add versioning policy	Prevents future breaking changes from being ambiguous.
Export OpenAPI spec	Supports SDK generation and security review.

Error Handling Recommendations

Area	Recommendation
Validation errors	Return structured field-level messages.
Authentication errors	Avoid leaking user existence or token internals.
Database errors	Log server-side details and return safe client messages.
Telemetry ingestion	Return accepted/rejected status with correlation IDs.
AI endpoints	Distinguish provider errors, fallback responses, and policy blocks.

Validation Recommendations

Area	Recommendation
IDs	Validate UUIDs consistently.
CIDR input	Limit scan size and validate allowed ranges.
RF measurements	Validate numeric bounds for RSSI, SNR, noise floor, packet loss, latency.
Logs	Enforce max payload size and severity enums.
Automation	Validate action permissions and require approval for risky actions.

Backend Security Recommendations

Risk	Recommendation
Weak JWT secret	Use a generated production secret and rotate safely.
Missing RBAC	Add roles, permissions, and organization isolation tests.

Risk	Recommendation
Exposed APIs	Add rate limiting, request size limits, and stricter CORS.
Telemetry abuse	Require agent identity, signed payloads, and replay protection.
AI misuse	Apply prompt injection controls and evidence-bound responses.

10. Frontend Documentation

The frontend is a Next.js 14 application written in TypeScript. It is the primary operator interface for the platform and provides pages for dashboard operations, assets, alerts, incidents, logs, security, automation, discovery, wireless links, diagnostics, radio devices, and field measurements.

UX Goals

Goal	Description
Operational clarity	Present dense network/security data in a scannable way.
Low friction	Minimize clicks required for common investigation workflows.
Consistent navigation	Use stable sidebar navigation and predictable page layouts.
Professional trust	Maintain serious visual tone suitable for NOC/SOC environments.
Field usability	Support technicians recording wireless measurements quickly.

Frontend Elements

Element	Expected Behavior
Dashboard UI	Summary cards, alerts, recent status, navigation into core modules.
Sidebar navigation	Persistent module access with clear labels.
Operational cards	Compact metrics, state counts, and urgent indicators.
Empty states	Explain absence of data without implying system failure.
Forms	Validate input, show clear errors, protect destructive actions.
Tables	Sortable/filterable lists for assets, logs, alerts, incidents, devices.
Wireless pages	Display RF metrics, diagnostics, maintenance history, and field actions.
Security pages	Expose rules, IOCs, risk states, and future detections.
Automation pages	Display playbooks, actions, and approval status.

Frontend Recommendations

Area	Recommendation
Loading states	Add skeleton rows/cards for all API-backed pages.
Error states	Show actionable messages and retry controls.
Empty states	Provide contextual next actions, such as “Add asset” or “Run scan”.
Form validation	Use shared schemas or generated API types where possible.

Area	Recommendation
Responsive layout	Ensure field technician pages work on tablets and small laptops.
Accessibility	Maintain keyboard navigation, focus states, semantic headings, and contrast.
Design system	Standardize badges, severities, status colors, tables, forms, and modals.

11. Database Documentation

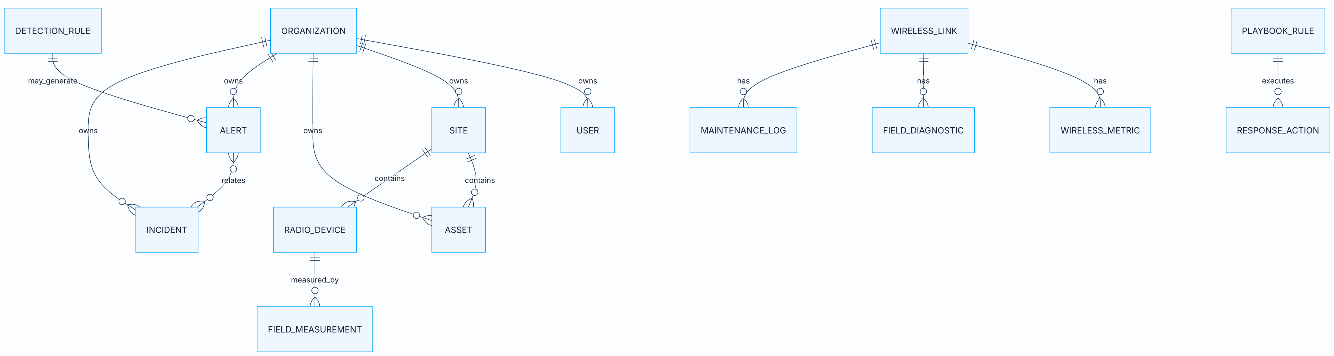
PostgreSQL 16 is the primary data store. It stores operational entities, security entities, wireless diagnostics data, user and organization context, and automation records.

Main Entities

Entity	Purpose
Organization	Tenant or administrative owner of records.
User	Authenticated platform user.
Site	Physical or logical location.
Asset	Network device, server, endpoint, or managed infrastructure item.
Alert	Operational or security condition requiring attention.
Incident	Investigation workflow, usually grouping related alerts.
LogEntry	Event/log record for observability and security review.
DetectionRule	Security rule definition.
IndicatorOfCompromise	IOC record for threat detection.
RadioDevice	Physical wireless radio device.
WirelessLink	PTP/PTMP link record.
WirelessMetric	Time-based RF/quality measurement.
FieldMeasurement	Technician measurement record.
FieldDiagnostic	Diagnostic record tied to a wireless link.
MaintenanceLog	Maintenance history for wireless links.
PlaybookRule	Automation rule/playbook definition.
ResponseAction	Recorded automation action.

Conceptual ERD

ARCHITECTURE DIAGRAM



12. Edge Agent Documentation

The Python edge agent is intended to collect telemetry close to the monitored environment. It can poll devices, normalize measurements, and send data to the backend.

Current Purpose

Function	Description
Telemetry collection	Collect measurements from local devices and radios.
Wireless metrics	Poll RSSI, SNR, CCQ, and related RF indicators where vendor support exists.
Local discovery	Future role for agent-based scans inside protected networks.
Backend communication	Push normalized metrics to the NetSentinel API.

Agent Security Requirements

Requirement	Recommendation
Agent identity	Assign each agent a unique ID and organization binding.
Agent authentication	Use signed tokens or mTLS rather than static development keys.
Payload integrity	Sign telemetry payloads and reject replayed timestamps.
Least privilege	Do not store broad device credentials in plain text.
Auditability	Log enrollment, configuration changes, and rejected payloads.

Retry and Offline Behavior

The mature agent should queue events locally when the backend is unavailable, retry with exponential backoff, preserve ordering where required, and report its own health. Offline queues should have retention limits, disk usage controls, and secure local storage.

13. Wireless Diagnostics

Outdoor wireless links require specialized diagnostic logic because link quality is affected by RF physics, antenna placement, weather, interference, channel planning, device configuration, and physical installation quality.

Wireless Link Types

Type	Description
Point-to-point	A dedicated link between two sites, commonly used for backhaul.
Point-to-multipoint	One access point communicates with multiple stations or CPEs.

RF and Quality Metrics

Metric	Meaning
RSSI	Received Signal Strength Indicator, commonly expressed in dBm. Less negative is stronger.
SNR	Signal-to-noise ratio. Higher values usually indicate cleaner signal quality.
Noise floor	Background RF noise level. A high noise floor can reduce usable signal quality.
CCQ	Client Connection Quality or link quality indicator, often reflecting retransmissions and stability.
Latency	Round-trip delay. High latency may indicate congestion, retransmissions, or poor RF quality.
Packet loss	Percentage of packets lost. Loss directly affects voice, video, and data reliability.
TX/RX capacity	Estimated transmit and receive throughput capacity.
Frequency	Operating frequency/channel. Poor frequency planning can cause interference.
Channel width	RF channel width. Wider channels can improve throughput but increase noise exposure.
Antenna alignment	Physical aiming quality between endpoints. Misalignment reduces RSSI and SNR.
Fresnel zone	Elliptical path around line-of-sight that should remain clear for strong propagation.
Interference	Competing RF energy that reduces signal quality and stability.
Modulation	Encoding scheme used by radio link. Lower modulation often means degraded link quality.
Link degradation	Decline in performance caused by RF, physical, configuration, or environmental problems.

Diagnostic Logic Table

Condition	Possible Cause	Severity	Recommended Technician Action
Weak RSSI	Antenna misalignment, obstruction, cable loss, wrong antenna gain, water ingress.	Medium to Critical	Inspect line of sight, verify Fresnel zone, check mounts, perform alignment sweep, inspect connectors.
Low SNR	Interference, high noise floor, weak signal, poor channel selection.	High	Run spectrum scan, change frequency, reduce channel width, improve alignment, verify shielding.
High noise floor	Congested spectrum, nearby transmitters, poor frequency reuse.	High	Scan spectrum at both ends, select cleaner channel, adjust channel width, coordinate frequency plan.
Low CCQ	Retransmissions, interference, firmware issue, poor modulation stability.	Medium to High	Check RF quality, review firmware, inspect error counters, test alternate channel.
High packet loss	RF instability, congestion, duplex mismatch, queue saturation, failing radio.	High	Run ping and throughput tests, check CPU/load, inspect RF metrics, isolate traffic congestion.
High latency	Congestion, retransmissions, overloaded radio, poor signal quality.	Medium to High	Review traffic load, queue settings, RF quality, and endpoint health.
Low capacity	Poor modulation, narrow channel, weak SNR, distance limits.	Medium	Review modulation rate, channel width, antenna gain, frequency, and regulatory limits.
Asymmetric throughput	One endpoint interference, TX power mismatch, antenna/cable issue on one side.	Medium	Test both directions, inspect each endpoint separately, compare local noise floor and RSSI.
Intermittent link	Loose mount, power issue, weather exposure, fading, cable fault, interference pattern.	Critical if service-affecting	Check power, grounding, cable ingress, mounting stability, and time-correlated RF trends.

Wireless Link Health Score Model

The proposed health score ranges from 0 to 100. It should be deterministic, explainable, and based on weighted RF and performance indicators.

Score Range	Meaning	Operational Interpretation
90-100	Excellent	Link is healthy with strong margin.
75-89	Good	Link is usable with minor monitoring recommendations.

Score Range	Meaning	Operational Interpretation
60-74	Degraded	Link needs review; performance risk is present.
40-59	Poor	Field intervention likely required.
0-39	Critical	Service-impacting failure or imminent failure likely.

Suggested scoring weights:

Metric	Weight
RSSI versus expected baseline	20
SNR	20
Noise floor	15
CCQ / retransmission quality	15
Packet loss	15
Latency	10
Capacity symmetry	5

> **IMPORTANT**
>
> RF thresholds must be vendor-aware and band-aware. A universal threshold can be useful for MVP triage, but production diagnostics should account for radio model, frequency band, link distance, antenna gain, modulation, channel width, and environmental profile.

14. Cybersecurity

NetSentinel AI aims to provide local-first security monitoring by connecting logs, security rules, indicators of compromise, alerts, incidents, and automation workflows.

Security Monitoring Goals

Goal	Description
Detect suspicious activity	Use rules, IOCs, logs, and telemetry to identify abnormal events.
Preserve local control	Keep sensitive operational data inside the local deployment by default.
Connect security to operations	Link security events to assets, incidents, and response workflows.
Support analyst review	Provide evidence, timeline, severity, and remediation context.

Risks and Mitigations

Risk	Impact	Mitigation
JWT risks	Token theft can enable account compromise.	Short-lived access tokens, refresh rotation, secure storage, revocation strategy.
Weak secrets	Attackers can forge tokens or access services.	Strong generated secrets, secret rotation, external secret store.
Exposed Docker ports	Local services may be reachable by unintended networks.	Bind to localhost where possible, firewall, reverse proxy, TLS.
Telemetry ingestion abuse	Attackers can flood or poison metrics.	Agent authentication, signed payloads, rate limits, schema validation.
AI prompt injection	Malicious logs or prompts may influence AI output.	Prompt isolation, evidence filtering, policy guardrails, source citations.
Data leakage	Sensitive logs or topology data may be exposed.	RBAC, audit logs, encryption, redaction, least privilege.
Missing rate limiting	APIs can be abused or overloaded.	Add per-user and per-agent rate limits.
Missing RBAC	Users may access data outside their role.	Implement roles, permissions, and organization isolation tests.
Missing tests	Regressions can compromise reliability.	Add unit, integration, API, security, and E2E tests.
Missing audit logs	Security actions lack traceability.	Record authentication, configuration, automation, and privileged actions.

SECURITY NOTE

Local-first does not automatically mean secure. It reduces cloud exposure, but the local deployment still requires secrets management, least privilege, secure configuration, and regular updates.

15. Deployment Documentation

Local Deployment

The MVP runtime is Docker Compose. The stack includes PostgreSQL, Redis, backend, frontend, worker, and edge-agent services.

Commands

```
cp .env.example .env
`docker compose up --build -d`
docker compose exec backend python -m app.seed
./Launch_NetSentinel.sh
```

Local Services

Service	URL / Port	Purpose
Dashboard	`http://localhost:3000`	Next.js dashboard UI.
Backend API	`http://localhost:8000`	FastAPI backend.
Swagger Docs	`http://localhost:8000/docs`	Interactive API documentation.
PostgreSQL	`localhost:5432`	Primary database.
Redis	`localhost:6379`	Cache and queue backend.

Docker Services

Compose Service	Image / Build	Purpose
postgres	`postgres:16-alpine`	Main relational database.
redis	`redis:7-alpine`	Cache and queue.
backend	`./backend`	FastAPI application.
frontend	`./frontend`	Next.js application.
worker	`./backend`	ARQ background worker.
edge-agent	`./edge-agent`	Telemetry and wireless polling agent.

Startup Checklist

Check	Expected Result
`.env` exists	Environment variables loaded.
Docker is running	Compose can start containers.

Check	Expected Result
Ports free	3000, 8000, 5432, and 6379 are available.
Database healthy	PostgreSQL healthcheck passes.
Redis healthy	Redis responds to ping.
Backend healthy	`/health` returns healthy status.
Frontend reachable	Dashboard loads at port 3000.
Seed data optional	Demo data available after seed command.

Troubleshooting Checklist

Symptom	Suggested Action
Port already in use	Stop conflicting service or change Compose ports.
Docker permission denied	Use Docker group configuration or authorized `pkexec` workflow.
Backend cannot connect to DB	Check `DATABASE_URL`, service name, and PostgreSQL health.
Frontend cannot call API	Check `NEXT_PUBLIC_API_URL` and CORS origins.
Desktop window opens early	Wait for Docker build and service readiness.
AI returns fallback	Configure `GEMINI_API_KEY` if live provider behavior is required.

Backup Recommendations

Asset	Recommendation
PostgreSQL data	Schedule `pg_dump` backups and test restores.
Environment secrets	Store outside Git and back up securely.
Uploaded reports	Store in versioned object/file storage if uploads are added.
Configuration	Export rules, IOCs, playbooks, and agent configs.

Production Hardening Recommendations

Area	Recommendation
Network exposure	Do not expose database or Redis publicly.
TLS	Terminate HTTPS through a reverse proxy.
Secrets	Use a secret manager or secure environment injection.
Authentication	Add RBAC, audit logs, token revocation, and secure cookie strategy.
Containers	Run non-root, pin images, scan dependencies, set resource limits.
Database	Use backups, migration discipline, least-privilege DB users.

16. Desktop Client

Electron is used to provide a local desktop experience for operators. Instead of requiring operators to remember URLs or manually open a browser, the desktop shell launches the local dashboard in a native window.

Advantages

Advantage	Description
Familiar application model	Operators can open NetSentinel AI from the Linux application menu.
Local workstation feel	Supports NOC/SOC workstation workflows.
Controlled entry point	Launcher can start services and open the dashboard.
Future system integration	Can support tray status, notifications, and local service health.

Future Improvements

Feature	Value
Tray icon	Quick status and launch controls.
Auto-start	Start monitoring environment after login.
Local status indicator	Show backend/frontend/database readiness.
Update system	Support packaged updates.
Offline mode	Show cached status when services are unavailable.
Packaged installers	Linux and Windows deployment packages.

17. AI Assistant

The AI assistant is intended to help operators understand incidents, alerts, wireless diagnostics, logs, and remediation options. It should act as an explanation and decision-support layer, not an autonomous authority.

Use Cases

Use Case	Description
Incident explanation	Summarize what happened and why it matters.
Wireless diagnostics	Explain RF degradation and suggest field actions.
Log summarization	Identify patterns and important events in logs.
Root cause analysis	Generate hypotheses based on internal evidence.
Suggested remediation	Propose next steps while requiring human approval.

AI Risks

Risk	Description
Hallucination	AI may invent causes or remediation steps.
Prompt injection	Malicious logs or user prompts may manipulate output.
Sensitive data exposure	Internal logs and topology may contain confidential information.
Overtrust	Operators may follow AI recommendations without verification.

Mitigations

Mitigation	Description
Evidence-based answers	Tie output to internal records and observed metrics.
Source citations	Reference incident IDs, alert IDs, logs, and metrics used.
Confidence levels	Expose uncertainty and alternative explanations.
Human approval	Require approval before automation or remediation actions.
Redaction	Remove secrets and sensitive fields before model calls.

18. Automation

Automation should standardize recurring operational and security responses without removing operator control.

Automation Concepts

Concept	Description
Playbook	Rule or workflow describing when and how to respond.
Action	Recorded or executed response step.
Human approval	Approval gate before risky changes.
Incident response	Standard process for triage, containment, remediation, and closure.
Alert handling	Deduplication, escalation, assignment, and notification.

Future Automated Remediation

Potential actions include creating tickets, sending notifications, quarantining an asset, blocking an IP, restarting a service, scheduling a field visit, or generating a PDF incident report. High-risk actions must be approval-based and fully audited.

19. Current MVP Status

Area	Status	Notes
Docker Compose local runtime	Implemented	Compose defines PostgreSQL, Redis, backend, frontend, worker, and edge-agent.
FastAPI backend	Implemented	Main app and routers exist.
Next.js frontend	Implemented	Multiple operational pages exist.
PostgreSQL models	Implemented	Core models are present.
JWT authentication	Implemented	Register, login, refresh, and current user routes exist.
Asset inventory	Implemented	Backend routes and frontend page exist.
Alerts and incidents	Implemented	CRUD and stats endpoints exist.
Logs	Partially implemented	Log creation/listing exists; advanced ingestion/search planned.
Security rules and IOCs	Partially implemented	Basic routes exist; correlation and detection need expansion.
Automation	Early MVP	Playbooks/actions listing exists; execution workflow needs expansion.
Discovery	Partially implemented	Real ICMP scan service exists; broader discovery planned.
Wireless diagnostics	Partially implemented	Wireless entities and AI brief concepts exist; deterministic scoring needs completion.
Edge agent	Early MVP	Polling structure exists with demo targets.
Desktop launcher	Implemented	Electron shell and Linux launcher exist.
Tests	Missing / needs verification	No clear project test suite found in backend or frontend source.
Security hardening	Planned	RBAC, rate limiting, audit logs, and hardened deployment are needed.
Production readiness	Not ready	MVP foundation only.

20. Roadmap

v0.1 MVP Stabilization

Area	Goals
Features	Stabilize existing dashboard, APIs, seed data, desktop launcher, and Docker runtime.
Technical tasks	Add migrations discipline, API response consistency, shared frontend types, error handling.
Security tasks	Replace dev secrets, document local-only assumptions, validate auth flows.
Documentation tasks	Maintain README, install guide, API overview, and this foundation dossier.

v0.2 Real Data & Diagnostics

Area	Goals
Features	Expand discovery, real wireless metric ingestion, diagnostic reports, field workflows.
Technical tasks	Add deterministic wireless health score and trend analysis.
Security tasks	Authenticate telemetry agents and validate payloads.
Documentation tasks	Add wireless diagnostics guide and field technician runbooks.

v0.3 Security Hardening

Area	Goals
Features	Security dashboard improvements, rules, IOCs, incident correlation.
Technical tasks	Add rate limiting, audit logs, RBAC, organization isolation tests.
Security tasks	Dependency scanning, threat model, secure deployment profile.
Documentation tasks	Security operations guide and hardening guide.

v0.4 Edge Agent Expansion

Area	Goals
Features	Agent enrollment, SNMP profiles, vendor adapters, offline queue.
Technical tasks	Agent configuration sync, retry logic, health reporting.
Security tasks	Signed payloads, agent tokens, replay protection.

Area	Goals
Documentation tasks	Agent installation and troubleshooting manual.

v0.5 Automation & AI Copilot

Area	Goals
Features	AI evidence citations, automation approval workflows, remediation suggestions.
Technical tasks	Playbook execution engine, AI context retrieval, confidence scoring.
Security tasks	Prompt injection controls and automation audit trail.
Documentation tasks	AI usage policy and automation playbook guide.

v1.0 Production-Ready Release

Area	Goals
Features	Role-based dashboards, reports, notifications, backup/restore, compliance views.
Technical tasks	E2E testing, observability, performance testing, installer packages.
Security tasks	RBAC, SSO/OIDC option, hardened Docker/Kubernetes profile.
Documentation tasks	Admin guide, operator guide, API reference, deployment guide, security guide.

21. Testing Strategy

Test Type	Scope
Backend unit tests	Services, auth utilities, validation logic, diagnostic scoring.
Backend integration tests	Database operations, router behavior, auth-protected endpoints.
API tests	OpenAPI contract, expected errors, pagination, filtering, status codes.
Frontend component tests	Tables, forms, badges, cards, navigation, error states.
End-to-end tests	Login, asset creation, discovery scan, alert workflow, wireless measurement.
Security tests	JWT validation, organization isolation, rate limits, CORS, input abuse.
Docker startup tests	Compose startup, health checks, seed command, service connectivity.
Wireless diagnostic rule tests	Known metric inputs produce expected severity and technician actions.
Agent tests	Polling errors, retry behavior, backend transport, offline queue.
Regression tests	Protect previously fixed workflows from breaking.

MVP LIMITATION

A formal test suite was not clearly present in the checked source tree. Adding tests should be treated as a near-term engineering priority.

22. Security Hardening Checklist

Item	Status
Strong JWT secret configured outside Git	Required
Access and refresh token expiration reviewed	Required
RBAC implemented	Required before production
Organization isolation enforced and tested	Required before production
Input validation for all endpoints	Required
API rate limiting	Required
CORS restricted to trusted origins	Required
Docker ports not publicly exposed	Required
Secrets management process	Required
Audit logs for auth/config/automation	Required
Password hashing with secure parameters	Present concept; verify settings
Database backup and restore tested	Required
Dependency scanning	Required
API abuse protection	Required
Telemetry agent authentication	Required
AI endpoint protection and prompt injection mitigation	Required
Security headers through reverse proxy	Recommended
TLS termination	Required beyond local-only use
Log redaction	Recommended
Incident export controls	Recommended

23. Business and Product Positioning

NetSentinel AI targets the gap between lightweight device monitoring tools and large enterprise NMS/SIEM platforms. Many organizations need practical local visibility, incident workflows, wireless diagnostics, and security monitoring but cannot justify large enterprise systems or do not want full cloud dependence.

Market Need

Need	Explanation
Local infrastructure visibility	Organizations need to understand assets, alerts, incidents, and logs.
Wireless diagnostics	WISPs and public institutions often depend on outdoor radio links.
Affordable NOC/SOC tooling	Small and medium teams need structured workflows without enterprise overhead.
Data control	Local-first deployment can support privacy and operational control.

Differentiation

Differentiator	Value
Local-first architecture	Keeps operational data close to the organization.
Wireless RF focus	Goes beyond standard uptime monitoring.
Combined NOC/SOC model	Connects operations and security workflows.
Desktop launcher	Supports operator workstation usage.
AI-assisted explanation	Helps translate complex evidence into practical actions.

Enterprise Potential

Future commercial positioning could include community edition, professional support, enterprise licensing, managed appliances, SIEM integrations, vendor adapters, compliance reporting, and role-based operational dashboards.

24. Future Enterprise Features

Feature	Description
Multi-tenant RBAC	Fine-grained roles and permissions across organizations.
SNMP integrations	Standard polling for network infrastructure.
MikroTik integration	RouterOS and wireless device support.
Ubiquiti integration	UISP/airMAX/UniFi-oriented device support where feasible.
TP-Link CPE integration	Support for common wireless CPE deployments.
Syslog ingestion	UDP/TCP syslog collection and parsing.
NetFlow/sFlow	Network traffic flow visibility.
SIEM export	Export alerts/events to external SIEM tools.
PDF reports	Incident, wireless diagnostic, and executive reports.
Role-based dashboards	Different views for executives, analysts, admins, and technicians.
Notification channels	Email, Slack, Teams, webhooks, SMS.
SLA tracking	Availability and incident response metrics.
Compliance reports	SOC 2, ISO 27001, local policy reporting support.
Backup and restore	Admin-controlled backup lifecycle.
Offline edge collectors	Resilient local agents for remote sites.

25. Official Appendices

Appendix A: Glossary

Term	Meaning
NOC	Network Operations Center.
SOC	Security Operations Center.
RF	Radio Frequency.
RSSI	Received Signal Strength Indicator.
SNR	Signal-to-noise ratio.
CCQ	Connection quality indicator often used in wireless systems.
IOC	Indicator of Compromise.
JWT	JSON Web Token.
PTP	Point-to-point wireless link.
PTMP	Point-to-multipoint wireless network.
Fresnel zone	Area around wireless line-of-sight that should remain unobstructed.
ARQ	Async Redis Queue for Python background jobs.

Appendix B: API Reference Summary

Module	Summary
Auth	Register, login, refresh token, current user.
Organizations	CRUD for organization records.
Sites	CRUD for site records.
Assets	CRUD and stats for assets.
Alerts	CRUD and stats for alerts.
Incidents	CRUD and stats for incidents.
Logs	Create and list log entries.
Security	Manage detection rules and IOCs.
Automation	List playbooks and response actions.
Discovery	Run scans, list scans, list hosts, import hosts as assets.
Wireless	Manage antenna profiles, mounts, interfaces, links, metrics, diagnostics, maintenance.
Field Measurements	Manage field measurements and stats.

Module	Summary
Radio Devices	Manage radio devices and diagnostics.
AI Assistant	Chat, incident analysis, alert analysis, capabilities.
Telemetry	Accept telemetry metrics.

Appendix C: Environment Variables

Variable	Purpose	Example
`POSTGRES_USER`	Database user	`netsentinel`
`POSTGRES_PASSWORD`	Database password	`netsentinel_dev_password`
`POSTGRES_DB`	Database name	`netsentinel`
`DATABASE_URL`	Backend database connection	`postgresql+asyncpg://...`
`REDIS_URL`	Redis connection	`redis://redis:6379/0`
`SECRET_KEY`	JWT signing secret	Replace in non-local use
`JWT_ALGORITHM`	JWT algorithm	`HS256`
`ACCESS_TOKEN_EXPIRE_MINUTES`	Access token lifetime	`30`
`REFRESH_TOKEN_EXPIRE_DAYS`	Refresh token lifetime	`7`
`NEXT_PUBLIC_API_URL`	Frontend API base URL	`http://localhost:8000/api/v1`
`ENVIRONMENT`	Runtime mode	`development`
`DEBUG`	Debug mode	`true`
`GEMINI_API_KEY`	Optional AI provider key	Empty by default

Appendix D: Docker Services Table

Service	Container Name	Port
PostgreSQL	`netsentinel-postgres`	`5432`
Redis	`netsentinel-redis`	`6379`
Backend	`netsentinel-backend`	`8000`
Frontend	`netsentinel-frontend`	`3000`
Worker	`netsentinel-worker`	Internal
Edge Agent	`netsentinel-edge-agent`	Internal

Appendix E: Example Wireless Measurement

Field	Example
Link name	North Tower to City Hall

Field	Example
Type	Point-to-point
Frequency	5.8 GHz
Channel width	40 MHz
RSSI	-68 dBm
SNR	27 dB
Noise floor	-95 dBm
CCQ	92%
Latency	4 ms
Packet loss	0.2%
TX capacity	180 Mbps
RX capacity	165 Mbps
Technician notes	Alignment acceptable; monitor during peak traffic.

Appendix F: Example Diagnostic Report

Section	Example
Summary	Link is degraded but service remains usable.
Evidence	RSSI is 8 dB worse than expected baseline; CCQ dropped below 85%.
Likely cause	Minor antenna movement or partial Fresnel obstruction.
Severity	Medium
Recommended action	Inspect mounting stability, verify line of sight, run spectrum scan.

Appendix G: Example Incident Workflow

Step	Action
1	Alert generated for degraded wireless link.
2	Incident opened and assigned to network technician.
3	AI assistant summarizes likely RF cause.
4	Field technician records measurements.
5	Technician realigns antenna and records maintenance log.
6	Alert resolves and incident is closed with notes.

Appendix H: Example Security Rule

Field	Example
Name	Repeated failed login attempts
Condition	More than 10 failed logins from same source in 5 minutes
Severity	High
Action	Create alert and recommend investigation
Future automation	Temporarily block source IP after approval

Appendix I: Example Automation Playbook

Field	Example
Name	Wireless degradation response
Trigger	Wireless health score below 60
Conditions	Packet loss above 3% or SNR below 18 dB
Actions	Create incident, notify technician, generate field brief
Approval	Required before service-impacting action

Appendix J: Recommended Folder Structure

```
NetSentinel AI/
  backend/
    app/
      models/
      schemas/
      routers/
      services/
      security/
      ai/
      ingestion/
      worker/
    alembic/
  frontend/
    src/
      app/
      components/
      lib/
  edge-agent/
  desktop-client/
  docker/
  docs/
  shared/
```

Appendix K: Recommended Documentation Structure

Document	Purpose
`README.md`	Project overview and quick start.
`INSTALL.md`	Local and desktop installation guide.
`architecture.md`	System architecture and design rationale.
`SECURITY.md`	Security policy and deployment warnings.
`roadmap.md`	Product roadmap and planned phases.
`docs/API.md`	API reference summary and examples.
`docs/OPERATOR_GUIDE.md`	Network/security operator workflows.
`docs/WIRELESS_DIAGNOSTICS.md`	RF measurement and diagnosis guide.
`docs/DEPLOYMENT_HARDENING.md`	Production hardening recommendations.